

INDENG 162 Final Project: Olympic Venue Selection, Scheduling, and Transportation Optimization

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Linear Programming and Network Flows

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Executive Summary:

Effective planning for the LA28 Olympic Games requires balancing venue costs, scheduling constraints, and overall operational efficiency. In this project, we developed a Mixed Integer Programming model to determine the most cost effective way to assign sports venues while satisfying all eligibility and capacity constraints. Using AMPL, we were able to solve for the optimal set of venues to open and the assignment of sports to those venues. Based on our results, the minimum total fixed cost is \$1,800,000. This was achieved by selecting a subset of venues that can host all 15 sports while minimizing unnecessary expenses. We then extended the model to incorporate things such as scheduling constraints, ticket revenue, and transportation options, allowing us to better reflect real-world constraints. These additions helped evaluate trade-offs between cost, capacity, and accessibility. Overall, our model provides a structured and data-driven approach to optimizing Olympic planning decisions, ensuring all events are hosted efficiently while minimizing total cost.

In **Task 1**, we started by building a baseline plan to host the Olympic events and required that each of the 15 sports be assigned to exactly one eligible venue. The model recommends opening five venues (V2, V4, V5, V7, & V8) for a fee of \$1,800,000. This layout successfully covers all 15 sports and represents the cheapest way to achieve basic event scheduling without exceeding any venue capacity.

In **Task 2**, we helped introduce a three-week scheduling dimension where each sport must run in exactly one week, and no venue can host more than one sport per week. Sports that require multiple sessions must occupy different venues during the same week. The model adjusts the scheduling to help meet these constraints while also keeping the base fixed cost as low as possible, resulting in a new minimum total cost of \$2,850,000.

In **Task 3**, we incorporated ticket sales, where each expected ticket sold generates \$10 in revenue to help offset the venue utilization costs. The tickets sold for each event are limited by either the sports demands or the venue's seating capacity. With this constraint, this changes the strategy to incentivize the model to select larger, slightly more expensive venues for high-demand sports to maximize overall profit. This resulted in a minimum cost of \$250,000.

In **Task 4**, we introduce optional bus lines to boost ticket sales by connecting two or three venues for a fixed cost of \$20,000 per network. The model recommends opening three bus networks, which helps generate enough cross-venue ticket sales to bring the net cost down to -\$44,800. At this stage, the total revenue completely exceeds the operational cost.

In **Task 5**, we performed a sensitivity analysis by running the Task 4 model under three different scenarios: ticket dropping to \$5, rising to \$15, and overall demand dropping by 10%. We find that the baseline strategy is highly resilient. Lowering the ticket price or the demand results in the same eight venues being selected. Across all three scenarios, the model opens the same eight venues; only the bus network configuration changes in response to ticket value shifts. The minimum costs corresponding to each scenario are \$1,424,100, \$1,520,400, and \$228,200, respectively.

Background:

As Operations Research consultants for the LA28 Olympic Games, we are tasked with determining the most cost-effective way to organize venue usage while ensuring that all sports are successfully hosted. We aim to solve the following key problem:

- Which venues should be opened and which sports should be assigned to each venue, while minimizing total operational cost.

This problem is important because venue activation and scheduling decisions significantly impact overall cost and efficiency for the Olympic Games.

The LA28 Committee has identified 10 potential venues across the Greater Los Angeles area and 15 Olympic sports that must be hosted. Each venue has a fixed cost to operate, a seating capacity, and a limit on how many sports it can host. Additionally, not all venues are capable of hosting every sport, so eligibility constraints must be taken into account when assigning sports venues.

The data used to solve this problem includes information on venue costs, capacities, and the maximum number of sports, as well as sport-specific requirements such as demand and eligibility. This data is used to determine feasible assignments and evaluate trade-offs between cost and capacity.

In order to solve this problem, we formulate a Mixed Integer Programming Model consisting of decision variables, parameters, an objective function, and constraints. We then implement this model in AMPL to compute the optimal solution. Using AMPL, we determine the most efficient set of venues to open and assign each sport to an eligible venue while minimizing total cost.

Modeling:

Venue Selection and Sport Assignment

This model determines which Olympic venues should be opened and which sports should be assigned to each venue while minimizing total fixed cost. A venue can either be opened and used to host sports or remain closed to avoid unnecessary cost. If a venue is opened, it can only host sports for which it is eligible. Later tasks add additional limits on the number of sports a venue can host and the weeks in which each venue is available.

Constraints:

1. Every Sport is Assigned to One Venue

Each sport must be assigned to exactly one venue. This is the basic requirement that ensures all Olympic sports are hosted.

2. Venue Eligibility

A sport can only be assigned to a venue if that venue is physically capable of hosting it. This prevents invalid assignments, such as assigning a sport to a location that does not have the correct facilities.

3. Open Venue Requirement

A sport can only be assigned to a venue if that venue is open. This connects the assignment decision to the venue selection decision.

4. Cost Minimization

The model minimizes the total fixed cost of all opened venues. This allows us to find the cheapest set of venues that can still host every sport.

Extensions in Later Tasks:

5. Maximum Sports Per Venue

In later tasks, each venue can only host up to its allowed number of distinct sports over the course of the Games. This prevents the model from overloading one venue beyond its scheduling or facility limit.

6. Multi-Week Scheduling

For sports that require multiple sessions, the model is extended so that these events can be scheduled across multiple venue week slots. A venue can host at most one sport per week. In addition, if a venue can host k sports, it is available during weeks 1 through k .

7. Ticket Revenue

Ticket sales are added to reduce the total net cost. The number of tickets sold depends on the smaller value between the sport demand and the venue capacity.

8. Bus Network Decisions

Bus routes are added as optimal decisions. These routes increase ticket sales by allowing spectators to attend multiple events in the same week, but they also add a fixed operating cost.

Mathematical Formulation (Task 1): The optimal cost is \$1,800,000 with 5 venues opened

The problem is formulated as a Mixed Integer Problem (MIP) with binary variables representing venue selection and sport assignments. The objective minimizes total fixed venue costs subject to assignment and eligibility. The full formulation is provided in Appendix A.

Task 1

$$y_j = \begin{cases} 1 & \text{if venue } j \text{ is open (assoc. w/ 1 time cost)} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{ij} = \text{if sport } i \text{ assigned to venue } j$$

$$a_{ij} = \text{if venue } j \text{ is eligible to host sport } i$$

$$i = 1, \dots, 15 \quad \text{sports}$$

$$j = 1, \dots, 10 \quad \text{venues}$$

$$\min \sum_{j=1}^{10} c_j y_j \quad \text{min. cost of opening venues}$$

$$\text{s.t.} \quad \sum_{j=1}^{10} x_{ij} = 1 \quad i = 1, \dots, 15 \quad \text{each sport assigned to exactly 1 venue}$$

$$x_{ij} \leq a_{ij} \cdot y_j \quad \forall i, j \quad \text{eligibility + open venue constraint}$$

$$\sum_{i=1}^{15} x_{ij} \leq k_j \cdot y_j \quad \forall j \quad \text{max. sports per venue}$$

$$x_{ij} \in \{0, 1\} \quad \text{binary constraints}$$

$$y_j \in \{0, 1\}$$

Mathematical Formulation (Task 2): The Optimal cost increases to \$2,850,000 with 8 venues opened

This model extended the base problem by adding a time dimension for multi-week scheduling. Binary variables y_j indicate whether a venue is open, $x_{i,j,t}$ assign sports to venues and weeks, and $w_{i,t}$ ensure each sport is scheduled in exactly one week. The objective minimizes total venue cost, while constraints enforce scheduling feasibility, venue eligibility, and limit each venue to at most one sport per week. The full formulation is provided in Appendix B.

task 2

- ↳ update problem w/ sub events
- ↳ venue can only host 1 sport per week

$$y_j = \begin{cases} 1 & \text{if venue } j \text{ is open (assoc. w/ 1 time cost)} \\ 0 & \text{otherwise} \end{cases}$$

$x_{i,j,t}$ = if sport i assigned to venue j during week t

$a_{i,j}$ = if venue j is eligible to host sport i

$w_{i,t}$ = if sport i is scheduled in week t

$i = 1, \dots, 15$ sports

$j = 1, \dots, 10$ venues

$$\min \sum_{j=1}^{10} c_j y_j \quad \text{min. cost of opening venues}$$

st.

$$\sum_t w_{i,t} = 1 \quad \text{each sport scheduled within exactly 1 week}$$

$$\sum_{j=1}^{10} x_{i,j,t} = R_i \cdot w_{i,t} \quad \text{during the week } t \text{ that sport } i \text{ is scheduled, it must occupy } R_i \text{ venues}$$

$$x_{i,j,t} \leq a_{i,j} \cdot y_j \quad \forall i,j,t \quad \text{eligibility + open venue constraint}$$

$$\sum_i \sum_t x_{i,j,t} \leq K_j \cdot y_j \quad \forall j \quad \text{max. sports per venue}$$

$$\sum_{i=1}^{15} x_{i,j,t} \leq 1 \quad \text{1 sport per venue per week}$$

$$x_{i,j,t} \in \{0,1\} \quad \text{binary constraints}$$

$$y_j \in \{0,1\}$$

Mathematical Formulation (Task 3): The net cost decreases to \$250,000 with 260,000 tickets sold

This model extended Task 2 by incorporating sales into the objective. Binary variables y_j , $x_{i,j,k}$, $w_{i,t}$ represent venue selection, sport assignments, and scheduling, while continuous variables $s_{i,j,t}$ represent ticket sales. The objective minimizes total venue cost while subtracting ticket revenue. Constraints ensure that ticket sales are limited by both sport demand and venue capacity, along with maintaining all scheduling and assignment constraints from previous tasks. The full formulation is provided in Appendix C.

task 3

↳ account for ticket sales

* ticket sales = $\min(\text{sport demand } D_i, \text{venue capacity})$

$y_j = \begin{cases} 1 & \text{if venue } j \text{ is open} \\ 0 & \text{otherwise} \end{cases}$ (assoc. w/ 1 time cost)

$x_{i,j,t}$ = if sport i assigned to venue j

$a_{i,j}$ = if venue j is eligible to host sport i

$w_{i,t}$ = if sport i is scheduled in week t

$s_{i,j}$ = ticket sales for venue j and sport i in week t

$i = 1, \dots, 15$ sports

$j = 1, \dots, 10$ venues

min $\sum_{j=1}^{10} c_j y_j - 10 \sum_i \sum_j s_{i,j}$ min. cost of opening venues - profit

st.

$\sum_t w_{i,t} = 1$ each sport scheduled within exactly 1 week

$\sum_{j=1}^{10} x_{i,j,t} = R_i \cdot w_{i,t}$ during the week t that sport i is scheduled, it must occupy R_i venues

$x_{i,j} \leq a_{i,j} \cdot y_j \quad \forall i,j,t$ eligibility + open venue constraint

$\sum_{i=1}^{15} x_{i,j,t} \leq K_j \cdot y_j \quad \forall j$ max. sports per venue

$\sum_{i=1}^{15} x_{i,j,t} \leq 1$ 1 sport per venue per week

$s_{i,j} \leq D_i \cdot \sum_t x_{i,j,t}$
 $s_{i,j} \leq C_j \cdot \sum_t x_{i,j,t}$ } ticket sales demand / capacity cap

$x_{i,j} \in \{0,1\}$ binary constraints

$y_j \in \{0,1\}$

$s_{i,j} \geq 0$ nonnegativity

Mathematical Formulation (Tasks 4 & 5):

This model extends task 3 by incorporating transportation decisions. Binary variables represent whether bus networks are open, while also making sure that each venue belongs to at most one network. The objective adds a fixed bus network cost and a bus bonus revenue contribution.

To keep the model strictly linear, the formulation introduces auxiliary binary variables z to help capture the "AND" logic between active bus networks and scheduled venues, along with continuous variables u and Big M constraints. These ensure the cross-venue ticket revenue boost is accurately calculated based only on the attendance of sports actively playing at connected venues. The original constraints from Task 3 are retained, with additional constraints requiring that buses only connect open venues. For task 5, the identical model is solved under three sensitivity scenarios that vary by either the ticket profit (ticket value = 5 in scenario 1 and ticket value = 15 in scenario 2) or the sport demand parameter (scaled to 90% of the original in scenario 3).

task 4

↳ buses

decision variables

$$y_i = \begin{cases} 1 & \text{if venue } i \text{ is open} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{ijt} = \begin{cases} 1 & \text{if sport } j \text{ is sched. in venue } i \text{ in week } t \\ 0 & \text{otherwise} \end{cases}$$

$$w_{jt} = \begin{cases} 1 & \text{sport } j \text{ runs in week } t \\ 0 & \text{otherwise} \end{cases}$$

$$s_{ijt} = \text{tickets sold for sport } j \text{ at venue } i \text{ during week } t$$

$$bus\ 2_{ik} = \begin{cases} 1 & \text{build 2 venue network betwn. venues } i \text{ + } k \\ 0 & \text{otherwise} \end{cases}$$

$$bus\ 3_{ikl} = \begin{cases} 1 & \text{build 3 venue network betwn. venues } i, k, l \\ 0 & \text{otherwise} \end{cases}$$

parameters

c_i = venue opening cost (in thousands of \$)

C_i = venue i capacity (in thousands of seats)

K_i = max distinct sports at venue i

$a_{ij} = 1$ if venue i can host sport j

D_j = demand for sport j (in thousands of fix/session)

R_j = required sessions for sport j

objective: minimizing the cost

$$\min \sum_i c_i y_i + 20 \left(\sum_i \sum_k bus\ 2_{ik} + \sum_i \sum_k \sum_l bus\ 3_{ikl} \right) - 10 \sum_j \sum_t s_{ijt} - 0.10 \sum_{(i,j,t) \in \text{venue pairs } (i,k)} (bus\ 2_{ik} \text{ active in week } t) - 0.14 \cdot 10 \sum_{(i,k,l) \in \text{venue triples}} (\alpha_{it} + \alpha_{kt} + \alpha_{lt}) \cdot (bus\ 3_{ikl} \text{ all active})$$

venue cost bus build cost per network ticket revenue

2. venue bonus 3. venue bonus - all 3 active

(partial active triple terms)

$$\begin{aligned}
\text{st } & \alpha_{it} = \sum_{j \in \text{Sports}} \min(D_j, C_i) x_{ijt} \quad \forall i, t \quad \text{attendance at venue } i \text{ in week } t \\
& \sum_t w_{jt} = 1 \quad \forall j \\
& \sum_i x_{ijt} = R_j w_{jt} \quad \forall j, t \quad \left. \begin{array}{l} \sum_t w_{jt} = 1 \quad \forall j \\ \sum_i x_{ijt} = R_j w_{jt} \quad \forall j, t \end{array} \right\} \text{sport scheduling} \\
& x_{ijt} \leq a_{ij} y_i \quad \forall i, j, t \\
& \sum_{j,t} x_{ijt} \leq K_i y_i \quad \forall i \\
& \sum_j x_{ijt} \leq 1 \quad \forall i, t \\
& x_{ijt} = 0 \quad \forall i, j, t \text{ with } t > K_i \quad \left. \begin{array}{l} \sum_{j,t} x_{ijt} \leq K_i y_i \quad \forall i \\ \sum_j x_{ijt} \leq 1 \quad \forall i, t \\ x_{ijt} = 0 \quad \forall i, j, t \text{ with } t > K_i \end{array} \right\} \text{venue rules} \\
& s_{ijt} \leq D_j x_{ijt} \quad \forall i, j, t \\
& s_{ijt} \leq C_i x_{ijt} \quad \forall i, j, t \quad \left. \begin{array}{l} s_{ijt} \leq D_j x_{ijt} \quad \forall i, j, t \\ s_{ijt} \leq C_i x_{ijt} \quad \forall i, j, t \end{array} \right\} \text{ticket caps} \\
& \sum_{(i,k) \ni i} \text{bus } z_{ik} + \sum_{(i,k,l) \ni i} \text{bus } z_{ikl} \leq 1 \quad \forall i \\
& \text{bus } z_{ik} \leq y_i, \text{bus } z_{ik} \leq y_k \\
& \text{bus } z_{ikl} \leq y_i, \text{bus } z_{ikl} \leq y_k, \text{bus } z_{ikl} \leq y_l \quad \left. \begin{array}{l} \sum_{(i,k) \ni i} \text{bus } z_{ik} + \sum_{(i,k,l) \ni i} \text{bus } z_{ikl} \leq 1 \quad \forall i \\ \text{bus } z_{ik} \leq y_i, \text{bus } z_{ik} \leq y_k \\ \text{bus } z_{ikl} \leq y_i, \text{bus } z_{ikl} \leq y_k, \text{bus } z_{ikl} \leq y_l \end{array} \right\} \text{bus network rules} \\
& y_i, x_{ijt}, w_{jt}, \text{bus } z_{ik}, \text{bus } z_{ikl} \in \{0,1\} \quad \text{binary constr.} \\
& s_{ijt} \geq 0 \quad \text{non-negativity}
\end{aligned}$$

★ this model is simplified, linearization constraints are in the AMPL code. the code solves the problem using auxiliary z indicator variables + a continuous variables

Results and Analysis:

Task 1: Venue Selection and Sports Assignment

After running our Mixed Integer Programming formulation through AMPL, we found that the optimal solution minimized the total fixed cost of opening venues to \$1,800,000. This model selects a subset of venues that can host all 15 Olympic sports while satisfying the basic Task 1 requirements: every sport must be assigned to exactly one open and eligible venue. Since Task 1 only uses fixed venue costs and the venue-sport eligibility matrix, this baseline solution focuses on finding the lowest-cost feasible set of venues before adding scheduling, capacity, revenue, and transportation considerations in later tasks.

The venues chosen to open are:

V2, V4, V5, V7, and V8, while V1, V3, V6, V9, and V10 remain closed. This indicates that the model can achieve full coverage of all sports using fewer venues than before, leading to a lower overall cost. By avoiding unnecessary or higher cost venues, the solution demonstrates strong cost efficiency.

This assignment of sports to venues is summarized below:

Venue	Assigned Sports
V2	S1, S4, S11, S14
V4	S5, S9
V5	S2, S3, S10
V7	S6, S7, S15
V8	S8, S12, S13

From this assignment, we observe that each open venue is used to cover several eligible sports. This helps reduce the need to open additional venues and lowers the total fixed cost. Additionally, all sports are assigned to exactly one eligible venue, satisfying the fundamental constraints of the model.

Hosting the Olympic Games involves significant financial planning, as infrastructure and operational costs must be carefully managed to ensure long-term economic benefits. Studies show that the economic impact of the Games depends heavily on efficient allocation of resources and minimizing unnecessary investments¹. This aligns with our model, which selects the minimum number of venues required to host all sports while maintaining feasibility.

In addition, research on sports facility optimization emphasizes that maximizing venue utilization and reducing underused capacity are critical for improving financial performance and

¹ International Olympic Committee. Economic Benefits of Hosting the Olympic Games, 2023.
<https://www.olympics.com/ioc/becoming-an-olympic-games-host/economic-benefits-of-hosting-the-olympic-games>cost-effective

operational efficiency². The results of our model reflect this principle, as each selected venue is used efficiently to host multiple sports while satisfying all Task 1 eligibility and assignment requirements.

Overall, Task 1 established a highly efficient baseline solution that minimizes total cost while ensuring all sports are feasibly assigned. This optimized venue selection will serve as the foundation for the more complex scheduling and revenue considerations introduced in later tasks.

Task 2: Multi-Week Scheduling Formulation

² Sports Facilities Companies. *Comprehensive Approach to Sports Facility Optimization*.
https://sportsfacilities.com/sf_services/management/sports-facility-optimization/

After running the multi-week scheduling model in AMPL, we found that the optimal total fixed cost was \$2,850,000. This is higher than the simple venue assignment model because Task 2 introduces additional scheduling constraints. In this version, each sport must be scheduled in exactly one week, and each venue can host at most one sport per week. In addition, the model uses each venue's max_sports parameter as a limit on the weeks in which that venue is available, so venues with lower availability cannot be scheduled in later weeks.

The model selected the following venues to open: **V1, V2, V3, V6, V7, V8, V9 and V10, while V4 and V5 remain closed.** This shows that the added scheduling requirements forced the model to open more venues to avoid conflicts across weeks.

The distribution of sports across weeks and venues is summarized below:

Week	Sports Scheduled	Venue Assignments
Week 1	S3, S6, S7, S8, S9, S11, S14	S3-V3/V9, S6-V7, S7-V6, S8-V8, S9-V10, S11-V1, S14-V2
Week 2	S1, S2, S5, S12, S13, S15	S1-V1/V2, S2-V6, S5-V3/V9, S12-V10, S13-V8, S15-V7
Week 3	S4, S10	S4-V1/V2, S10-V3

The main difference from Task 1 is that sports requiring multiple sessions now need multiple venue slots within the same week. For example, S1, S3, S4, and S5 each require two venue assignments. Because each venue can only host one sport per week, the model must distribute events across more venues and time periods, reducing overall flexibility.

This result reflects real-world scheduling challenges. In practice, sports scheduling becomes significantly more complex as the number of constraints increases, including limits on venue usage, fairness considerations, and coordination across multiple locations³. Additionally, large scale tournaments scheduling must account for factors such as venue availability, travel coordination, and preventing scheduling conflicts, which requires solving thousands of constraints simultaneously⁴.

³ SimpleRose. Why Sports Scheduling is So Hard and How It Can Be Done Better, 2026
<https://simplerose.com/blog/sports-scheduling/>

⁴ Fastbreak. Top AI Scheduling Software for Multi-Venue Sports Tournaments 2025.
<https://www.fastbreak.ai/blog/ai-scheduling-tools-multi-venue-sports-tournaments>

Task 3: Ticket Sales

After incorporating ticket sales into the AMPL model, the optimal net cost decreased to \$250,000. The model continues to open **V1, V2, V3, V5, V7, V8, V9, and V10, while V4 and V6 remain closed.** Before including ticket revenue, the fixed cost of these venues was \$2,850,000; however, the ticket sales generated approximately \$2,600,000 in revenue, reducing the final net cost significantly. The ticket sales introduce a continuous variable $s_{i,j,t}$, representing the number of tickets sold for sport i at venue j during week t . Ticket sales are constrained by both the sport's expected demand and the venue's seating capacity, meaning the number of tickets sold is limited by the smaller of the two values. The distribution of ticket sales across weeks is shown below:

Week	Sports Scheduled	Total Tickets Sold
Week 1	S1, S2, S5, S8, S13, S15	109,000
Week 2	S3, S6, S7, S9, S11, S12, S14	75,000
Week 3	S4, S10	76,000
Total		260,000

The highest ticket sales occur in Week 1, primarily because Athletics (S1) is assigned to both V1 and V2, selling 30,000 tickets at each venue. Similarly, Soccer (S4) generates high revenue in Week 3, selling 32,000 tickets at both V1 and V2.

In practice, ticket sales are constrained by both demand and venue capacity, as events cannot sell more tickets than available seating.⁵ Additionally, event revenue is influenced by demand-based pricing strategies, where higher demand leads to increased ticket sales and overall revenue.⁶ More broadly, ticket pricing and sales are governed by basic supply and demand principles, where limited venue capacity and high popularity increase the value of tickets.⁷

⁵ Joshi, Praggya. *Demystifying Event Ticket Pricing Strategies*, 2024 <https://www.eventcombo.com/a/1077/demystifying-event-ticket-pricing-strategies-to-find-the-bes>

⁶ Dynamic Ticket Pricing and Revenue Optimization, 2025. <https://www.freshtix.com/blog/maximizing-revenue-with-dynamic-ticket-pricing-strategies>

⁷ Chirco, Jackie. *Economics of Concert Ticket Pricing*, 2024. <https://tseentertainment.com/event-ticketing-understanding-the-economics-of-concert-ticket-pricing/>

Overall, Task 3 demonstrates that incorporating ticket revenue significantly changes the objective of the model. Instead of only minimizing venue costs, the model now balances cost with revenue potential. High-demand sports and larger-capacity venues become more valuable, as they generate greater revenue and help offset the operational costs of hosting the Olympics.

Task 4: Bus Network Decisions

For Task 4, we extended the ticket sales model by adding bus network decisions. The model considers both two-venue and three-venue bus networks. To prevent duplicate combinations, the set PAIRS and TRIPLES are ordered using the numerical venue index.

The main new decision variables are bus2 and bus3. The variable bus2 equals 1 if a two-venue bus network is opened, while bus 3 equals 1 if a three-venue bus network is opened. We also add an active variable, where $active_{j,t} = 1$ if a venue j hosts any sport during the week t .

Because the model needs to apply ticket boosts only when a bus network is open and the connected venues are active in the same week, we introduce auxiliary variables such as $z2$, $z3jk$, $z3jl$, $z3kl$, and $z3all$. These variables represent “AND” logic between bus network decisions and weekly venue activity.

For example, $z2_{j,k,t}$ equals 1 only if the two venue network between venues j k is open and both venues are active during the week t . This is enforced using linearization constraints. The upper bounds $z2 \leq bus2$, $z2 \leq active_j$, and $z2 \leq active_k$ force $z2$ to be 0 if any condition is false. The lower bound $z2 \geq bus2 + active_j + active_k - 2$ forces $z2$ to equal 1 when all three conditions are true. The objective adds a fixed cost of 20 for each selected bus network and subtracts additional ticket revenue based on 10% attendance boosts for active connected venue pairs. For two-venue networks, the model applies the required 10% cross-venue ticket boost when both connected venues are active in the same week. For three-venue networks, the specification requires spectators at each active venue to generate 7% additional purchases for each of the other two venues when all three venues are active, for a total 14% boost per venue. To model this linearly, the objective first uses pairwise 10% terms and then subtracts a 6% adjustment when all three venues are active. This converts the pairwise 20% effect into the required 14% total effect. If only two venues in a three-venue network are active, the model behaves like a two-venue network, applying the 10% boost between the two active venues.

The optimal solution builds three bus networks: a 2-venue network connecting V8–V10, and two 3-venue networks connecting V1–V2–V3 and V5–V7–V9.

After extending the model to include bus network decisions, we introduce transportation links between venues to improve accessibility and increase ticket sales. These networks allow spectators to travel between venues within the same week, creating additional demand and increasing overall revenue.

Bus Network Decision Summary:

Network Type	Description	Effect on Model
2 Venue Network (bus 2)	Connect two venues	Increases ticket sales if both venues are active in the same week

3 Venue Network (bus 3)	Connects three venues	Creates a larger accessibility boost across multiple venues
Active Variable	Indicates if a venue hosts an event in a given week	Ensures networks only apply when venues are in use
Auxiliary Variable (z)	Capture “AND” logic between bus and activity	Ensure the revenue boost only applies when all conditions are met

The model incorporates both transportation costs and revenue increases from improved accessibility. Bus networks are only selected when the additional ticket revenue outweighs their fixed operating cost. The venue selection remains consistent with Task 3, with V1, V2, V3, V5, V7, V8, V9, and V10 open, while V4 and V6 remain closed. This indicates that transportation improvements enhance the efficiency of the existing system rather than requiring additional venues. Bus networks are most effective when connecting high-demand sports and high-capacity venues occurring in the same week, as these combinations maximize the revenue boost. The model avoids unnecessary connections, ensuring that transportation investments are only made when they provide a net benefit. In real-world event planning, transportation plays a critical role in overall success by improving accessibility and ensuring that attendees can move efficiently between venues.⁸ Large-scale events also significantly increase travel demand and congestion, making coordinated transportation systems essential for maintaining efficient operations and maximizing attendance.⁹ By incorporating bus networks, the model reflects these real-world dynamics, increasing effective demand and improving overall system performance.

Task 5: Sensitivity Analysis

In Task 5, we analyze how sensitive our optimal solution is to changes in ticket value and demand. The baseline model from Task 4 assumes a ticket value of \$10 per ticket and full demand. We consider three alternative scenarios: reducing ticket value to \$5, increasing it to \$15, and reducing demand by 10%.

Scenario A: Ticket Value = \$5

When the ticket value is reduced to \$5, the optimal cost becomes +\$1,424.1K. The model selects the following venues: V1, V2, V3, V5, V7, V8, V9, and V10, while V4 and V6 remain closed.

⁸ Lockerby Transportation Group, The Role of Transportation in Event Planning Success, 2025.
<https://www.lockerbytransportation.com/bridging-the-gap-the-role-of-transportation-in-event-planning-success>

⁹ U.S. Department of Transportation. Planned Special Events: Economic and Congestion Impacts
https://ops.fhwa.dot.gov/publications/fhwahop08022/exec_sum.htm

Compared to the baseline, the solutions favor lower-cost venues, since ticket revenue is less valuable. As a result, the model avoids opening higher cost venues like V4, even though they have larger capacities. The overall schedule remains feasible, but the system prioritized minimizing fixed cost over maximizing revenue.

Scenario B: Ticket Value = \$15

When the ticket value increases to \$15, the optimal cost significantly decreases to -\$1,520.4K, indicating much higher profitability. The selected venues remain V1, V2, V3, V5, V7, V8, V9, and V10, while V4 and V6 remain closed. This shows that the venue selection is robust, even when ticket revenue becomes more valuable. The main changes occur in the bus network configuration rather than in the venue selection.

Scenario C: Demand Reduced by 10%

When demand is reduced by 10%, the optimal cost becomes +\$228.2K. The selected venues are identical to Scenario A: V1, V2, V3, V5, V7, V8, V9, and V10. Although total ticket sales decrease (from about 260K to 236K), the venue selection does not change. This indicates that a moderate reduction in demand is not enough to justify switching to higher cost venues, and the model tends to favor the same cost-efficient configuration.

Comparison of Facility Selection:

Across the three scenarios, the venue selection remains unchanged. Scenarios A, B, and C all open V1, V2, V3, V5, V7, V8, V9, and V10. Therefore, the main differences across scenarios appear in the bus network choices and the resulting net cost, not in the facility selection.

- Scenario A keeps both 3-venue networks but drops the V8–V10 2-venue network.
- Scenario B keeps the same network structure as the baseline but swaps V9 and V10's roles: V9 joins the 2-venue network with V8, while V10 joins the V5–V7 triple.
- Scenario C matches the Task 4 Baseline exactly.

This would indicate that venue selection is robust to changes in ticket value and demand within the range; the eligibility constraints, capacity limits, and session requirements help drive together a stable optimal venue set.

Sub-Optimality Across Scenarios

We also evaluate how each solution performs when applied to other scenarios:

- Solutions from Scenarios A and C perform identically across all cases, showing strong robustness
- The solution from Scenario B is less robust, performing worse when the ticket value decreases or the demand drops

Specifically:

- Scenario B's solution incurs up to +57.8K higher cost when applied to Scenario A
- Scenarios A and C are only about +26.6K worse when applied to Scenario B

Key Insights:

Overall, Task 5 shows that the model is highly sensitive to ticket value in terms of total net cost, but the selected venue set remains stable across the tested scenarios. Lower ticket values reduce revenue and increase net cost, while higher ticket values increase profitability. However, because the eligibility constraints, session requirements, and scheduling limits strongly restrict the feasible venue choices, the same eight venues are selected in all three sensitivity scenarios. The main changes occur in the bus network configuration rather than in the venue selection. This suggests that venue selection is relatively robust, while transportation decisions are more sensitive to changes in ticket value and demand.

Table 1: Facilities Comparison

Scenario	Venues Opened
A (\$5)	V1, V2, V3, V5, V7, V8, V9, V10
B(\$15)	V1, V2, V3, V5, V7, V8, V9, V10
C(-10% demand)	V1, V2, V3, V5, V7, V8, V9, V10

Table 2: Cost Comparison

Scenario	Optimal Cost (\$K)
A (\$5)	1424.1
B(\$15)	-1520.4
C(-10%)	228.2

Appendix:

Appendix A: AMPL Model Task 1

.mod file for task 1

```
set VENUES;
set SPORTS;
set WEEKS;

param cost      {VENUES};
param capacity  {VENUES};
param max_sports {VENUES};
param eligible  {SPORTS, VENUES};
param demand    {SPORTS};
param sessions  {SPORTS};
param venue_order {VENUES} >= 0;

var y {VENUES} binary;
var x {SPORTS, VENUES} binary;

minimize TotalCost:
    sum {j in VENUES} cost[j] * y[j];

subject to

assign_sport {i in SPORTS}:
    sum {j in VENUES} x[i,j] = 1;

eligibility {i in SPORTS, j in VENUES}:
    x[i,j] <= eligible[i,j] * y[j];
```

Appendix B: AMPL Model Task 2

.mod file for task 2

```
set VENUES;
set SPORTS;
set WEEKS;

param cost {VENUES};
param capacity {VENUES};
param max_sports {VENUES};
param eligible {SPORTS, VENUES};
param sessions {SPORTS};
param demand {SPORTS};
param venue_order {VENUES} >= 0;

var y {VENUES} binary;
var x {SPORTS, VENUES, WEEKS} binary;
var w {SPORTS, WEEKS} binary;

minimize Total_Cost:
    sum {j in VENUES} cost[j] * y[j];

subject to

one_week {i in SPORTS}:
    sum {t in WEEKS} w[i,t] = 1;

sessions_constraint {i in SPORTS, t in WEEKS}:
    sum {j in VENUES} x[i,j,t] = sessions[i] * w[i,t];

eligibility {i in SPORTS, j in VENUES, t in WEEKS}:
    x[i,j,t] <= eligible[i,j] * y[j];

max_sports_limit {j in VENUES}:
    sum {i in SPORTS, t in WEEKS} x[i,j,t] <= max_sports[j] * y[j];

one_sport_per_week {j in VENUES, t in WEEKS}:
    sum {i in SPORTS} x[i,j,t] <= 1;

week_availability {i in SPORTS, j in VENUES, t in WEEKS: t > max_sports[j]}:
    x[i,j,t] = 0;
```

Appendix C: AMPL Model Task 3

.mod file for task 3

```
set VENUES;
set SPORTS;
set WEEKS;

param cost {VENUES};
param capacity {VENUES};
param max_sports {VENUES};
param eligible {SPORTS, VENUES};
param sessions {SPORTS};
param demand {SPORTS};
param venue_order {VENUES} >= 0;

var y {VENUES} binary;
var x {SPORTS, VENUES, WEEKS} binary;
var w {SPORTS, WEEKS} binary;
var s {SPORTS, VENUES, WEEKS} >= 0;

minimize Total_Cost:
    sum {j in VENUES} cost[j] * y[j]
    - 10 * sum {i in SPORTS, j in VENUES, t in WEEKS} s[i,j,t];

subject to

one_week {i in SPORTS}:
    sum {t in WEEKS} w[i,t] = 1;

sessions_constraint {i in SPORTS, t in WEEKS}:
    sum {j in VENUES} x[i,j,t] = sessions[i] * w[i,t];

eligibility {i in SPORTS, j in VENUES, t in WEEKS}:
    x[i,j,t] <= eligible[i,j] * y[j];

max_sports_limit {j in VENUES}:
    sum {i in SPORTS, t in WEEKS} x[i,j,t] <= max_sports[j] * y[j];

one_sport_per_week {j in VENUES, t in WEEKS}:
    sum {i in SPORTS} x[i,j,t] <= 1;
```

```
week_availability {i in SPORTS, j in VENUES, t in WEEKS: t > max_sports[j]}:  
    x[i,j,t] = 0;  
  
ticket_demand {i in SPORTS, j in VENUES, t in WEEKS}:  
    s[i,j,t] <= demand[i] * x[i,j,t];  
  
ticket_capacity {i in SPORTS, j in VENUES, t in WEEKS}:  
    s[i,j,t] <= capacity[j] * x[i,j,t];
```

Appendix D: AMPL Model Task 4

.mod file for task 4

```
set VENUES;
set SPORTS;
set WEEKS;

param venue_order {VENUES} >= 0;

set PAIRS := {j in VENUES, k in VENUES: venue_order[j] < venue_order[k]};
set TRIPLES := {j in VENUES, k in VENUES, l in VENUES:
    venue_order[j] < venue_order[k] and venue_order[k] <
venue_order[l]};

param cost {VENUES} >= 0;
param capacity {VENUES} >= 0;
param max_sports {VENUES} >= 0;
param eligible {SPORTS, VENUES} >= 0;
param sessions {SPORTS} >= 0;
param demand {SPORTS} >= 0;

param ticket_value := 10;
param bus_cost := 20;

param tps {i in SPORTS, j in VENUES} :=
    if demand[i] <= capacity[j] then demand[i] else capacity[j];

# Big-M per venue: max possible attendance from any eligible sport
param Mv {j in VENUES} := max {i in SPORTS} tps[i,j];

var y {VENUES} binary;
var x {SPORTS, VENUES, WEEKS} binary;
var w {SPORTS, WEEKS} binary;
var s {SPORTS, VENUES, WEEKS} >= 0;

var bus2 {(j,k) in PAIRS} binary;
var bus3 {(j,k,l) in TRIPLES} binary;
var active {VENUES, WEEKS} binary;

var z2 {(j,k) in PAIRS, t in WEEKS} binary;
var z3jk {(j,k,l) in TRIPLES, t in WEEKS} binary;
var z3jl {(j,k,l) in TRIPLES, t in WEEKS} binary;
```

```

var z3kl {(j,k,l) in TRIPLES, t in WEEKS} binary;
var z3all {(j,k,l) in TRIPLES, t in WEEKS} binary;

# Attendance at venue j during week t
var attend {VENUES, WEEKS} >= 0;

# Per-pair: attend gated by z2
var u2j {(j,k) in PAIRS, t in WEEKS} >= 0;
var u2k {(j,k) in PAIRS, t in WEEKS} >= 0;

# Per-triple: attend gated by each z3 indicator
var u3jk_j {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3jk_k {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3jl_j {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3jl_l {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3kl_k {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3kl_l {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3all_j {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3all_k {(j,k,l) in TRIPLES, t in WEEKS} >= 0;
var u3all_l {(j,k,l) in TRIPLES, t in WEEKS} >= 0;

minimize Total_Cost:
    (sum {j in VENUES} cost[j] * y[j])
    + bus_cost * (sum {(j,k) in PAIRS} bus2[j,k]
        + sum {(j,k,l) in TRIPLES} bus3[j,k,l])
    - ticket_value * (sum {i in SPORTS, j in VENUES, t in WEEKS} s[i,j,t])
    - ticket_value * 0.10 * (sum {(j,k) in PAIRS, t in WEEKS} (u2j[j,k,t] + u2k[j,k,t]))
    - ticket_value * (sum {(j,k,l) in TRIPLES, t in WEEKS} (
        0.10 * (u3jk_j[j,k,l,t] + u3jk_k[j,k,l,t])
        + 0.10 * (u3jl_j[j,k,l,t] + u3jl_l[j,k,l,t])
        + 0.10 * (u3kl_k[j,k,l,t] + u3kl_l[j,k,l,t])
        - 0.06 * (u3all_j[j,k,l,t] + u3all_k[j,k,l,t] + u3all_l[j,k,l,t])
    ));

subject to

# Define attendance: equals scheduled sport's tps, or 0
attend_def {j in VENUES, t in WEEKS}:
    attend[j,t] = sum {i in SPORTS} tps[i,j] * x[i,j,t];

active_eq {j in VENUES, t in WEEKS}:
    active[j,t] = sum {i in SPORTS} x[i,j,t];

```



```

one_network {j in VENUES}:
    (sum {(j2,k) in PAIRS:      j2 = j or k = j}          bus2[j2,k])
    + (sum {(j2,k,l) in TRIPLES: j2 = j or k = j or l = j} bus3[j2,k,l]) <= 1;

z2_ub1 {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] <= bus2[j,k];
z2_ub2 {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] <= active[j,t];
z2_ub3 {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] <= active[k,t];
z2_lb  {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] >= bus2[j,k] + active[j,t] +
active[k,t] - 2;

z3jk_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: z3jk[j,k,l,t] <= bus3[j,k,l];
z3jk_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: z3jk[j,k,l,t] <= active[j,t];
z3jk_ub3 {(j,k,l) in TRIPLES, t in WEEKS}: z3jk[j,k,l,t] <= active[k,t];
z3jk_lb  {(j,k,l) in TRIPLES, t in WEEKS}: z3jk[j,k,l,t] >= bus3[j,k,l] + active[j,t]
+ active[k,t] - 2;

z3jl_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: z3jl[j,k,l,t] <= bus3[j,k,l];
z3jl_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: z3jl[j,k,l,t] <= active[j,t];
z3jl_ub3 {(j,k,l) in TRIPLES, t in WEEKS}: z3jl[j,k,l,t] <= active[l,t];
z3jl_lb  {(j,k,l) in TRIPLES, t in WEEKS}: z3jl[j,k,l,t] >= bus3[j,k,l] + active[j,t]
+ active[l,t] - 2;

z3kl_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: z3kl[j,k,l,t] <= bus3[j,k,l];
z3kl_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: z3kl[j,k,l,t] <= active[k,t];
z3kl_ub3 {(j,k,l) in TRIPLES, t in WEEKS}: z3kl[j,k,l,t] <= active[l,t];
z3kl_lb  {(j,k,l) in TRIPLES, t in WEEKS}: z3kl[j,k,l,t] >= bus3[j,k,l] + active[k,t]
+ active[l,t] - 2;

z3all_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: z3all[j,k,l,t] <= bus3[j,k,l];
z3all_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: z3all[j,k,l,t] <= active[j,t];
z3all_ub3 {(j,k,l) in TRIPLES, t in WEEKS}: z3all[j,k,l,t] <= active[k,t];
z3all_ub4 {(j,k,l) in TRIPLES, t in WEEKS}: z3all[j,k,l,t] <= active[l,t];
z3all_lb  {(j,k,l) in TRIPLES, t in WEEKS}: z3all[j,k,l,t] >= bus3[j,k,l]
+ active[j,t] + active[k,t] + active[l,t] - 3;

# Big-M linearization: u = attend when z = 1, else 0
u2j_ub1 {(j,k) in PAIRS, t in WEEKS}: u2j[j,k,t] <= attend[j,t];
u2j_ub2 {(j,k) in PAIRS, t in WEEKS}: u2j[j,k,t] <= Mv[j] * z2[j,k,t];
u2j_lb  {(j,k) in PAIRS, t in WEEKS}: u2j[j,k,t] >= attend[j,t] - Mv[j] * (1 -
z2[j,k,t]);

```

```

u2k_ub1 {(j,k) in PAIRS, t in WEEKS}: u2k[j,k,t] <= attend[k,t];
u2k_ub2 {(j,k) in PAIRS, t in WEEKS}: u2k[j,k,t] <= Mv[k] * z2[j,k,t];
u2k_lb {(j,k) in PAIRS, t in WEEKS}: u2k[j,k,t] >= attend[k,t] - Mv[k] * (1 -
z2[j,k,t]);

u3jk_j_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_j[j,k,l,t] <= attend[j,t];
u3jk_j_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_j[j,k,l,t] <= Mv[j] * z3jk[j,k,l,t];
u3jk_j_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_j[j,k,l,t] >= attend[j,t] - Mv[j] *
(1 - z3jk[j,k,l,t]);

u3jk_k_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_k[j,k,l,t] <= attend[k,t];
u3jk_k_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_k[j,k,l,t] <= Mv[k] * z3jk[j,k,l,t];
u3jk_k_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_k[j,k,l,t] >= attend[k,t] - Mv[k] *
(1 - z3jk[j,k,l,t]);

u3jl_j_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_j[j,k,l,t] <= attend[j,t];
u3jl_j_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_j[j,k,l,t] <= Mv[j] * z3jl[j,k,l,t];
u3jl_j_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_j[j,k,l,t] >= attend[j,t] - Mv[j] *
(1 - z3jl[j,k,l,t]);

u3jl_l_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_l[j,k,l,t] <= attend[l,t];
u3jl_l_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_l[j,k,l,t] <= Mv[l] * z3jl[j,k,l,t];
u3jl_l_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_l[j,k,l,t] >= attend[l,t] - Mv[l] *
(1 - z3jl[j,k,l,t]);

u3kl_k_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_k[j,k,l,t] <= attend[k,t];
u3kl_k_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_k[j,k,l,t] <= Mv[k] * z3kl[j,k,l,t];
u3kl_k_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_k[j,k,l,t] >= attend[k,t] - Mv[k] *
(1 - z3kl[j,k,l,t]);

u3kl_l_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_l[j,k,l,t] <= attend[l,t];
u3kl_l_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_l[j,k,l,t] <= Mv[l] * z3kl[j,k,l,t];
u3kl_l_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_l[j,k,l,t] >= attend[l,t] - Mv[l] *
(1 - z3kl[j,k,l,t]);

u3all_j_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_j[j,k,l,t] <= attend[j,t];
u3all_j_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_j[j,k,l,t] <= Mv[j] *
z3all[j,k,l,t];
u3all_j_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3all_j[j,k,l,t] >= attend[j,t] - Mv[j]
* (1 - z3all[j,k,l,t]);

u3all_k_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_k[j,k,l,t] <= attend[k,t];

```

```

u3all_k_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_k[j,k,l,t] <= Mv[k] *
z3all[j,k,l,t];
u3all_k_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3all_k[j,k,l,t] >= attend[k,t] - Mv[k]
* (1 - z3all[j,k,l,t]);

u3all_l_ub1 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_l[j,k,l,t] <= attend[l,t];
u3all_l_ub2 {(j,k,l) in TRIPLES, t in WEEKS}: u3all_l[j,k,l,t] <= Mv[l] *
z3all[j,k,l,t];
u3all_l_lb {(j,k,l) in TRIPLES, t in WEEKS}: u3all_l[j,k,l,t] >= attend[l,t] - Mv[l]
* (1 - z3all[j,k,l,t]);

# Bus networks can only use opened venues
bus2_open_j {(j,k) in PAIRS}: bus2[j,k] <= y[j];
bus2_open_k {(j,k) in PAIRS}: bus2[j,k] <= y[k];

bus3_open_j {(j,k,l) in TRIPLES}: bus3[j,k,l] <= y[j];
bus3_open_k {(j,k,l) in TRIPLES}: bus3[j,k,l] <= y[k];
bus3_open_l {(j,k,l) in TRIPLES}: bus3[j,k,l] <= y[l];

# z indicators can't fire if the bus isn't built or the venue isn't active
# (tighter versions of the existing z2_lb)
z2_open_j {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] <= y[j];
z2_open_k {(j,k) in PAIRS, t in WEEKS}: z2[j,k,t] <= y[k];

# u variables capped by the actual attendance bound (tighter than Mv when y=0)
u2j_y {(j,k) in PAIRS, t in WEEKS}: u2j[j,k,t] <= Mv[j] * y[j];
u2k_y {(j,k) in PAIRS, t in WEEKS}: u2k[j,k,t] <= Mv[k] * y[k];

u3jk_j_y {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_j[j,k,l,t] <= Mv[j] * y[j];
u3jk_k_y {(j,k,l) in TRIPLES, t in WEEKS}: u3jk_k[j,k,l,t] <= Mv[k] * y[k];
u3jl_j_y {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_j[j,k,l,t] <= Mv[j] * y[j];
u3jl_l_y {(j,k,l) in TRIPLES, t in WEEKS}: u3jl_l[j,k,l,t] <= Mv[l] * y[l];
u3kl_k_y {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_k[j,k,l,t] <= Mv[k] * y[k];
u3kl_l_y {(j,k,l) in TRIPLES, t in WEEKS}: u3kl_l[j,k,l,t] <= Mv[l] * y[l];
u3all_j_y {(j,k,l) in TRIPLES, t in WEEKS}: u3all_j[j,k,l,t] <= Mv[j] * y[j];
u3all_k_y {(j,k,l) in TRIPLES, t in WEEKS}: u3all_k[j,k,l,t] <= Mv[k] * y[k];
u3all_l_y {(j,k,l) in TRIPLES, t in WEEKS}: u3all_l[j,k,l,t] <= Mv[l] * y[l];

# Original Task 2/3 constraints
one_week {i in SPORTS}: sum {t in WEEKS} w[i,t] = 1;

```

```

sessions_constraint {i in SPORTS, t in WEEKS}:
    sum {j in VENUES} x[i,j,t] = sessions[i] * w[i,t];

eligibility {i in SPORTS, j in VENUES, t in WEEKS}:
    x[i,j,t] <= eligible[i,j] * y[j];

max_sports_limit {j in VENUES}:
    sum {i in SPORTS, t in WEEKS} x[i,j,t] <= max_sports[j] * y[j];

one_sport_per_week {j in VENUES, t in WEEKS}:
    sum {i in SPORTS} x[i,j,t] <= 1;

week_availability {i in SPORTS, j in VENUES, t in WEEKS: t > max_sports[j]}:
    x[i,j,t] = 0;

ticket_demand {i in SPORTS, j in VENUES, t in WEEKS}:
    s[i,j,t] <= demand[i] * x[i,j,t];

ticket_capacity {i in SPORTS, j in VENUES, t in WEEKS}:
    s[i,j,t] <= capacity[j] * x[i,j,t];

```

Tasks 5a, 5b, and 5c use the same code as Task 4. The ticket prices and data files are altered based on the scenarios.

Appendix E: Data Files

Data file for tasks 1-5b

```

set VENUES := V1 V2 V3 V4 V5 V6 V7 V8 V9 V10;
set SPORTS := S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15;
set WEEKS := 1 2 3;

```

```

param cost :=
    V1 500  V2 500  V3 400  V4 400  V5 350
    V6 350  V7 300  V8 250  V9 300  V10 250;

```

```

param capacity :=
    V1 70  V2 90  V3 20  V4 27  V5 12
    V6 10  V7 11  V8 8  V9 17  V10 8;

```

param max_sports :=

V1 3 V2 3 V3 3 V4 2 V5 2
V6 2 V7 2 V8 2 V9 2 V10 2;

param venue_order :=

V1 1 V2 2 V3 3 V4 4 V5 5
V6 6 V7 7 V8 8 V9 9 V10 10;

param demand :=

S1 30 S2 15 S3 9 S4 32 S5 9
S6 8 S7 7 S8 5 S9 10 S10 12
S11 20 S12 6 S13 4 S14 8 S15 10;

param sessions :=

S1 2 S2 1 S3 2 S4 2 S5 2
S6 1 S7 1 S8 1 S9 1 S10 1
S11 1 S12 1 S13 1 S14 1 S15 1;

param eligible :

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	
S1	1	1	0	0	0	0	0	0	0	0	
S2	0	0	0	0	1	1	0	0	0	0	
S3	0	0	1	0	1	0	0	0	1	0	
S4	1	1	0	1	0	0	0	0	0	0	
S5	0	0	1	1	0	0	0	0	1	0	
S6	0	0	0	0	0	1	1	0	1	0	
S7	0	0	0	0	0	1	1	0	0	1	
S8	0	0	0	0	0	0	0	1	0	1	
S9	0	0	0	1	0	0	0	0	0	1	
S10	0	0	1	0	1	0	0	0	1	0	
S11	1	1	0	0	0	0	0	0	0	0	
S12	0	0	0	0	0	0	0	1	0	1	
S13	0	0	0	0	0	0	0	1	0	0	
S14	0	1	0	0	0	0	0	0	0	0	
S15	0	0	0	0	0	0	1	0	0	0	;

Data file for Task 5c

```
set VENUES := V1 V2 V3 V4 V5 V6 V7 V8 V9 V10;  
set SPORTS := S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15;  
set WEEKS := 1 2 3;
```

```
param cost :=  
  V1 500 V2 500 V3 400 V4 400 V5 350  
  V6 350 V7 300 V8 250 V9 300 V10 250;
```

```
param capacity :=  
  V1 70 V2 90 V3 20 V4 27 V5 12  
  V6 10 V7 11 V8 8 V9 17 V10 8;
```

```
param max_sports :=  
  V1 3 V2 3 V3 3 V4 2 V5 2  
  V6 2 V7 2 V8 2 V9 2 V10 2;
```

```
param venue_order :=  
  V1 1 V2 2 V3 3 V4 4 V5 5  
  V6 6 V7 7 V8 8 V9 9 V10 10;
```

```
param demand :=  
  S1 27 S2 13.5 S3 8.1 S4 28.8 S5 8.1  
  S6 7.2 S7 6.3 S8 4.5 S9 9 S10 10.8  
  S11 18 S12 5.4 S13 3.6 S14 7.2 S15 9;
```

```
param sessions :=  
  S1 2 S2 1 S3 2 S4 2 S5 2  
  S6 1 S7 1 S8 1 S9 1 S10 1  
  S11 1 S12 1 S13 1 S14 1 S15 1;
```

```
param eligible :  
  V1 V2 V3 V4 V5 V6 V7 V8 V9 V10 :=  
  S1 1 1 0 0 0 0 0 0 0 0  
  S2 0 0 0 0 1 1 0 0 0 0  
  S3 0 0 1 0 1 0 0 0 1 0  
  S4 1 1 0 1 0 0 0 0 0 0  
  S5 0 0 1 1 0 0 0 0 1 0  
  S6 0 0 0 0 0 1 1 0 1 0
```

S7	0	0	0	0	0	1	1	0	0	1
S8	0	0	0	0	0	0	0	1	0	1
S9	0	0	0	1	0	0	0	0	0	1
S10	0	0	1	0	1	0	0	0	1	0
S11	1	1	0	0	0	0	0	0	0	0
S12	0	0	0	0	0	0	0	1	0	1
S13	0	0	0	0	0	0	0	1	0	0
S14	0	1	0	0	0	0	0	0	0	0
S15	0	0	0	0	0	0	1	0	0	0;

Appendix F: Solver Outputs:

Task 1

```

HiGHS 1.11.0:   alg:parallel = on
    tech:threads = 4
    mip:gap = 0.01
    lim:time = 180
    pre:solve = on
HiGHS 1.11.0: optimal solution; objective 1800
0 simplex iterations
1 branching nodes

```

Total cost: \$1,800,000 (objective: 1800.00K)

Open venues:

```

V2
V4
V5
V7
V8
(total: 5 venues)

```

Schedule:

```

V2 hosts S1
V5 hosts S10
V2 hosts S11
V8 hosts S12
V8 hosts S13
V2 hosts S14
V7 hosts S15
V5 hosts S2

```

V5 hosts S3
V2 hosts S4
V4 hosts S5
V7 hosts S6
V7 hosts S7
V8 hosts S8
V4 hosts S9

1800

Task 2

```
HiGHS 1.11.0:  alg:parallel = on
                tech:threads = 4
                mip:gap = 0.01
                lim:time = 180
                pre:solve = on
HiGHS 1.11.0: optimal solution; objective 2850
41 simplex iterations
1 branching nodes
```

Total cost: \$2,850,000 (objective: 2850.00K)

Open venues:

V1
V10
V2
V3
V6
V7
V8
V9
(total: 8 venues)

Schedule:

Week 1:
V1 hosts S11
V10 hosts S9
V2 hosts S14
V3 hosts S3
V6 hosts S7
V7 hosts S6
V8 hosts S8
V9 hosts S3

Week 2:

V1 hosts S1
V10 hosts S12
V2 hosts S1
V3 hosts S5
V6 hosts S2
V7 hosts S15
V8 hosts S13
V9 hosts S5
Week 3:
V1 hosts S4
V2 hosts S4
V3 hosts S10

2850

Task 3

HiGHS 1.11.0: alg:parallel = on
tech:threads = 4
mip:gap = 0.01
lim:time = 180
pre:solve = on
HiGHS 1.11.0: optimal solution; objective 250
64 simplex iterations
1 branching nodes

Total cost: \$250,000 (objective: 250.00K)

Open venues:

V1
V10
V2
V3
V5
V7
V8
V9
(total: 8 venues)

Schedule:

Week 1:
V1 hosts S1
V10 hosts S8
V2 hosts S1
V3 hosts S5
V5 hosts S2
V7 hosts S15

V8 hosts S13
V9 hosts S5

Week 2:

V1 hosts S11
V10 hosts S9
V2 hosts S14
V3 hosts S3
V5 hosts S3
V7 hosts S7
V8 hosts S12
V9 hosts S6

Week 3:

V1 hosts S4
V2 hosts S4
V3 hosts S10

Tickets sold per sport:

S1: 60.0K tickets
S10: 12.0K tickets
S11: 20.0K tickets
S12: 6.0K tickets
S13: 4.0K tickets
S14: 8.0K tickets
S15: 10.0K tickets
S2: 12.0K tickets
S3: 18.0K tickets
S4: 64.0K tickets
S5: 18.0K tickets
S6: 8.0K tickets
S7: 7.0K tickets
S8: 5.0K tickets
S9: 8.0K tickets
TOTAL: 260.0K tickets (260,000 actual tickets)

250

Task 4

HiGHS 1.11.0: alg:parallel = on
tech:threads = 4
mip:gap = 0.01
lim:time = 180
pre:solve = on
Warning code 1 while solving with HiGHS
HiGHS 1.11.0: time limit, feasible solution; objective -44.8
138343 simplex iterations
122 branching nodes

absmipgap=187, relmipgap=4.17411

Total profit: \$44,800 (objective: -44.80K)

Open venues:

V1

V10

V2

V3

V5

V7

V8

V9

(total: 8 venues)

Bus networks:

2-venue: V8 <-> V10

3-venue: V1 <-> V2 <-> V3

3-venue: V5 <-> V7 <-> V9

Schedule:

Week 1:

V1 hosts S1

V10 hosts S9

V2 hosts S1

V3 hosts S3

V5 hosts S3

V7 hosts S7

V8 hosts S8

V9 hosts S6

Week 2:

V1 hosts S4

V10 hosts S12

V2 hosts S4

V3 hosts S5

V5 hosts S2

V7 hosts S15

V8 hosts S13

V9 hosts S5

Week 3:

V1 hosts S11

V2 hosts S14

V3 hosts S10

Tickets sold per sport:

S1: 60.0K tickets

S10: 12.0K tickets

S11: 20.0K tickets
S12: 6.0K tickets
S13: 4.0K tickets
S14: 8.0K tickets
S15: 10.0K tickets
S2: 12.0K tickets
S3: 18.0K tickets
S4: 64.0K tickets
S5: 18.0K tickets
S6: 8.0K tickets
S7: 7.0K tickets
S8: 5.0K tickets
S9: 8.0K tickets
TOTAL: 260.0K tickets (260,000 actual tickets)

-44.79999999999971

Task 5a

HiGHS 1.11.0: alg:parallel = on
tech:threads = 4
mip:gap = 0.01
lim:time = 180
pre:solve = on
HiGHS 1.11.0: optimal solution; objective 1424.1
37877 simplex iterations
3 branching nodes
absmipgap=13.7, relmipgap=0.00962011

Total cost: \$1,424,100 (objective: 1424.10K)

Open venues:

V1
V10
V2
V3
V5
V7
V8
V9
(total: 8 venues)

Bus networks:

3-venue: V1 <-> V2 <-> V3
3-venue: V5 <-> V7 <-> V9

Schedule:

Week 1:

V1 hosts S4
V10 hosts S9
V2 hosts S4
V3 hosts S5
V5 hosts S2
V7 hosts S15
V8 hosts S13
V9 hosts S5

Week 2:

V1 hosts S1
V10 hosts S8
V2 hosts S1
V3 hosts S3
V5 hosts S3
V7 hosts S7
V8 hosts S12
V9 hosts S6

Week 3:

V1 hosts S11
V2 hosts S14
V3 hosts S10

Tickets sold per sport:

S1: 60.0K tickets
S10: 12.0K tickets
S11: 20.0K tickets
S12: 6.0K tickets
S13: 4.0K tickets
S14: 8.0K tickets
S15: 10.0K tickets
S2: 12.0K tickets
S3: 18.0K tickets
S4: 64.0K tickets
S5: 18.0K tickets
S6: 8.0K tickets
S7: 7.0K tickets
S8: 5.0K tickets
S9: 8.0K tickets
TOTAL: 260.0K tickets (260,000 actual tickets)

1424.09999999999526

Task 5b

```
HiGHS 1.11.0:   alg:parallel = on
  tech:threads = 4
  mip:gap = 0.01
  lim:time = 180
  pre:solve = on
  Warning code 1 while solving with HiGHS
HiGHS 1.11.0: time limit, feasible solution; objective -1520.4
138814 simplex iterations
130 branching nodes
absmipgap=297.3, relmipgap=0.195541
```

Total profit: \$1,520,400 (objective: -1520.40K)

Open venues:

```
V1
V10
V2
V3
V5
V7
V8
V9
(total: 8 venues)
```

Bus networks:

```
2-venue: V8 <-> V9
3-venue: V1 <-> V2 <-> V3
3-venue: V5 <-> V7 <-> V10
```

Schedule:

Week 1:

```
V1 hosts S1
V10 hosts S9
V2 hosts S1
V3 hosts S3
V5 hosts S3
V7 hosts S7
V8 hosts S13
V9 hosts S6
```

Week 2:

```
V1 hosts S4
V10 hosts S12
V2 hosts S4
V3 hosts S5
V5 hosts S2
V7 hosts S15
V8 hosts S8
```

V9 hosts S5
Week 3:
V1 hosts S11
V2 hosts S14
V3 hosts S10

Tickets sold per sport:

S1: 60.0K tickets
S10: 12.0K tickets
S11: 20.0K tickets
S12: 6.0K tickets
S13: 4.0K tickets
S14: 8.0K tickets
S15: 10.0K tickets
S2: 12.0K tickets
S3: 18.0K tickets
S4: 64.0K tickets
S5: 18.0K tickets
S6: 8.0K tickets
S7: 7.0K tickets
S8: 5.0K tickets
S9: 8.0K tickets
TOTAL: 260.0K tickets (260,000 actual tickets)

-1520.40000000000005

Task 5c

HiGHS 1.11.0: alg:parallel = on
tech:threads = 4
mip:gap = 0.01
lim:time = 180
pre:solve = on
Warning code 1 while solving with HiGHS
HiGHS 1.11.0: time limit, feasible solution; objective 228.2
176421 simplex iterations
532 branching nodes
absmipgap=148.059, relmipgap=0.648813

Total cost: \$228,200 (objective: 228.20K)

Open venues:

V1
V10
V2

V3
V5
V7
V8
V9
(total: 8 venues)

Bus networks:

2-venue: V8 <-> V10
3-venue: V1 <-> V2 <-> V3
3-venue: V5 <-> V7 <-> V9

Schedule:

Week 1:

V1 hosts S4
V10 hosts S12
V2 hosts S4
V3 hosts S5
V5 hosts S2
V7 hosts S15
V8 hosts S8
V9 hosts S5

Week 2:

V1 hosts S1
V10 hosts S9
V2 hosts S1
V3 hosts S3
V5 hosts S3
V7 hosts S7
V8 hosts S13
V9 hosts S6

Week 3:

V1 hosts S11
V2 hosts S14
V3 hosts S10

Tickets sold per sport:

S1: 54.0K tickets
S10: 10.8K tickets
S11: 18.0K tickets
S12: 5.4K tickets
S13: 3.6K tickets
S14: 7.2K tickets
S15: 9.0K tickets
S2: 12.0K tickets
S3: 16.2K tickets
S4: 57.6K tickets

S5: 16.2K tickets

S6: 7.2K tickets

S7: 6.3K tickets

S8: 4.5K tickets

S9: 8.0K tickets

TOTAL: 236.0K tickets (236,000 actual tickets)

228.19999999992962